

birdclef-2026

4th Place Solution: BirdCLEF+ 2026

Rank	4
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BirdCLEF 2026 — 4th Place Solution

Thanks to all of our teammates! We're glad to see no shake-down — many public top-10 teams dropped significantly, and fortunately we had selected our highest private LB submission.

Overview

Our solution is a 5-model ensemble combining a Perch v2 bioacoustic foundation model, two custom-trained SED (Sound Event Detection) models, and two Perch-distilled SED models. We leveraged pretrained XC backbones from the BirdCLEF 2025 2nd place solution, Perch knowledge distillation, and multi-model blending with taxonomic post-processing.

Model Summary Table

	perch	v2s	v2s_2025_2nd	v2s_perch_distill	se26_perch_distill
Type	Foundation	SED	SED (OpenVINO)	SED + Distill	SED + Distill
Backbone	Perch v2	EfficientNetV2-S-IN21K	EfficientNetV2-S-IN21K	EfficientNetV2-S-IN21K	SeriesResNext26d-32x4d
Pretrain	Perch v2 (Google)	2025 2nd place XC pretrain	2025 2nd place XC pretrain	2025 2nd place XC pretrain	2021/22/23/24 data pretrained
Mel bins	—	128	128	256	256
Hop size	—	256	512	512	512
fmin / fmax	—	40 / 14000	20 / None	20 / 16000	20 / 16000
Mel power	—	2.5	2.0	2.0	2.0
Duration	5s	10s	5s	5s	5s
Batch size	—	256	64	64	64
Epochs	—	40	50	30	30
LR	—	5e-5	1e-4	5e-4	5e-4
Optimizer	—	AdamW	AdamW	AdamW	AdamW
Scheduler	—	CosineAnnealing	CosineWarmRestarts	Warmup+Cosine	Warmup+Cosine
Loss	—	Weighted BCE	FocalBCE	BCE	BCE
Balanced sampler	—	✓	✓	✓	✓

	perch	v2s	v2s_2025_2nd	v2s_perch_distill	se26_perch_distill
Mixup	—	Audio-level mix (p=0.5)	Half-mix (p=0.5)	Audio-level mix (p=0.5)	Audio-level mix (p=0.5)
SpecAug	—	✗	FreqMask + TimeMask	FreqMask + TimeMask	FreqMask + TimeMask
Random Filtering	—	✗	✓ (STFT-based)	✗	✗
Soundscapes (5x)	—	✓	✓	✓ (labeled SC)	✓ (labeled SC)
Pseudo labels	—	✗	✗	0.938 SED and 0.934 PerchDistill, 0.5 threshold	0.938 SED and 0.934 PerchDistill 0.5 threshold
Standalone LB	~0.928	~0.938	~0.930	~0.942	~0.942
Ensemble weight	0.40	0.182 (0.70 in SED)	0.078 (0.30 in SED)	0.085 (0.25 in PDENS)	0.255 (0.75 in PDENS)

Training Details

perch

We used the [Google Perch v2](#) bioacoustic foundation model directly without any fine-tuning. It serves as both a standalone predictor and a prototype ranker in the ensemble.

v2s

Our baseline SED model:

- **Backbone:** EfficientNetV2-S-IN21K with pretrained weights from the [BirdCLEF 2025 2nd place solution](#), fine-tuned on 2026 competition data with a 234-class head
- **Data:** Competition train + soundscapes (5× upsampled) + external data (iNat/XC, threshold=100 per species)
- **Augmentation:** Audio-level mixing (50%), frequency filter SpecAug disabled

we start with Starry's [BirdClef 2025 code](#). First of all, the 2021/22/23/24 data was used to pretrain the SED model (tf_efficientnetv2_b3 backbone) with BCE loss. It scored ~0.897. The parameters I used: mel_bin: 192 f_min/f_max: 40/140000 n_fft: 2048 win_length/hop_size: 1024/512 train/infer durations: 10/5s

public LB improvement journey (single model, Running time: ~20min)

Methods	Scored
Baseline	0.897

Methods	Scored
remove spectrum aug / mel mixup / upsample labeled soundscapes	0.910
use last year efv2s pretrained models	0.928
add time prior generated by claude code	0.935
hop_len -> 256, mel_bin -> 128, mel_power -> 2.5	0.938

v2s_2025_2nd

We reproduced the [2025 2nd-place SED architecture](#) with minimal modifications for 2026:

- **Backbone:** EfficientNetV2-S-IN21K, structure and pretrained weights both from 2025 2nd place
- **Loss:** Focal BCE ($\alpha=0.25$, $\gamma=2.0$)
- **Augmentation:** STFT-based random frequency filtering, frequency/time masking, half-proportion audio mixing
- **Converted to OpenVINO** IR format for CPU-efficient inference on Kaggle

v2s_perch_distill & se26_perch_distill (Perch Knowledge Distillation)

This pipeline is modified from [tuckerarrants' public code](#). And we do some slightly modification.

These models are trained to simultaneously classify bird species AND align with [Perch v2](#) embeddings:

Architecture:

- SED backbone (EFv2-S or SeriesResNext26d) + additional distillation projection head (Linear $\rightarrow$ 1536-dim Perch embedding space)
- Generalized Mean pooling (learnable exponent) for frequency aggregation
- Temporal attention + classification heads

Training:

public LB improvement journey (single model, Running time: ~20min)

Methods	Scored
baseline (efv2b3 backbone)	0.914
use last year efv2s pretrained models	0.934
0.934 PerchDistill model and 0.935 SED model as teacher to generate unlabeled soundscapes pseudo labels	0.942
modified to seresnext26d_32 $\times$ 4d backbone	0.942

Ensemble & Post-Processing

Ensemble Weights

Group	Models	Inner weight	Outer weight
perch	perch	—	0.40
SED blend	v2s	0.70	0.26 (total)
	v2s_2025_2nd	0.30	
PDENS blend	v2s_perch_distill	0.25	0.34 (total)
	se26_perch_distill	0.75	

Blend strategy: Rank-gating composite — perch serves as a species ranking prototype that gates SED confidence scores, preventing weaker SED models from introducing false positives in rare species.

Post-Processing Pipeline

1. **Genus smoothing** ($\alpha=0.12$): Species within the same taxonomic genus have probabilities smoothed toward each other — helps co-occurring related species
2. **Time-of-day prior** ($\alpha=1.0$): Per-species hourly activity distributions applied to all predictions
3. **Sonotype mirroring**: Synthetic sound variants share the max prediction of their group
4. **Fat-tail temporal smoothing**: Kernel radius=3, weights $\propto \text{distance}^{-1.5}$

Inference Pipeline (Kaggle CPU-4 Constraint)

All inference runs within Kaggle's 4 vCPU limit using a dynamic CPU-budget scheduler:

Budget: 4 vCPU total, 2 vCPU per lane

Timeline:

```
[0s]  perch (2 CPU) + v2s_perch_distill (2 CPU) → start together
[Xm]  perch done → v2s starts (2 CPU)
[Ym]  v2s_perch_distill done → v2s_2025_2nd starts (2 CPU)
[Zm]  v2s done → se26_perch_distill starts (overlaps v2s_2025_2nd tail)
[Wm]  All done → BLEND
```

v2s_2025_2nd is served via OpenVINO for CPU-efficient inference; perch via ONNX Runtime.

What Didn't Work

- **Self-trained XC pretraining:** We attempted to pretrain our own backbone on 814K XC recordings (~7,489 species) from scratch. Results were consistently worse than directly using the pretrained weights from the [2025 2nd place solution](#).

- **Standard pseudo-labeling:** Using ensembled SED model outputs as soft labels for soundscape segments did not improve performance. Only Perch-based pseudo labels (from the foundation model) led to meaningful gains.
- **AnuraSet external data:** Including frog recordings from AnuraSet caused a slight LB drop due to domain mismatch.
- **External data threshold >100 per species:** Marginal improvement vs. added noise.

There maybe something we lose. Feel free to ask any questions!

Supplemental Q&A

Ethan Kershner · 2026-06-04T12:07:22.583000

Great work & congratulations to the team! I am curious - were you guys able to figure out an approach to cross-validation? I am asking because I fell down quite a bit on the private vs. public leaderboard so am wondering how I could have possibly prevented this. I tried a lot of different cv approaches but none seemed to work well for me - local cv never seemed to correlate with gains on the public lb.

Starry · 2026-06-04T12:22:54.403000

What you concern is what I concern when I begin to participate in BirdClef series competetion at 2023 — I can not find a reliable cross validation strategy. The answer maybe a little bit disappointed: I still can not find a reliable cv strategy. I just train the model locally, submit it, get a score as the reference. Then tune my train stratagy and submit it again. In other words, public LB as CV. The only things I do to aviod overfitting to public LB is try to find some intuitive ideas instead of over-tuning hyperparameters.

chenwenqiang_001 · 2026-06-06T15:28:02.407000

Congratuations.Could you show more details abut v2s ? how to group? how to use focal and soundscapes? is mixup mixing within a batch? how to set the weights for weighted bce?

Starry · 2026-06-06T15:47:02.063000

Sorry could you please decribe more about group?? for focal and label soundscapes, I upsample the 4× labeled soundscapes and concatenate it with focal samples. for mixup, the answer is yes, batch mixup, the weight of bce is 1:19 for negative and positive samples.

chenwenqiang_001 · 2026-06-07T01:08:05.513000

Thank you for your patiently answering. I mean how to split into folds.do you use groupkfold or stratifiedkfold?Does v2s only use one fold ? about negative , do you mean nocall samples?You use extra data as negative samples?

Starry · 2026-06-08T02:20:14.103000

We use all data to train, use a small portation(10%) of train set as validataion set, which means the validtion set is meaningless. And for the negative, since the task is a multi-label classification task, so the negative is the prediction of output except the target class.